CYBERSECURITY PROJECT PRESENTATION

PHISH GUARD

Intelligent phishing detection & awareness platform

LIVE PROJECT phish-guard-rho-green.vercel.app

Pause • Check • Protect

TEAM MEMBERS

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Project overview

Phish Guard is a web-based cybersecurity platform that helps users recognize suspicious digital interactions before they become victims of scams.

URL SCANNING

Checks website links for suspicious patterns.

CALL CHECKING

Reviews caller claims for common scam warning signs.

SMS ANALYSIS

Looks for phishing and scam indicators in messages.

AWARENESS QUIZ

Helps users practise safer online decisions.

Problem statement & proposed solution

THE PROBLEM

Online scams often rely on urgency, impersonation and misleading links.

- Users may not recognize look-alike URLs.
- Fraudulent calls and messages can request OTPs or banking details.
- People need simple, accessible guidance.

THE SOLUTION

One platform to check, learn and make more informed decisions.

- Analyze URLs, calls and SMS content.
- Look up numbers using a local directory.
- Track scan activity and quiz performance.

A single platform, multiple protection tools

01 Dashboard

Overview of scans, risk categories and quiz analytics.

02 URL Scanner

Analyzes a website URL and presents a risk score with findings.

03 Call Checker

Evaluates caller statements for common scam indicators.

04 SMS Checker

Analyzes message text for suspicious language and requests.

05 Phone Lookup

Checks numbers against the local directory and saved history.

06 Quiz & History

Supports awareness learning and review of past activity.

Tools and technologies used

The application uses a browser interface, a server-side API and a cloud database.

FRONTEND

React.js and Vite build the interactive pages users see in their browser.

BACKEND

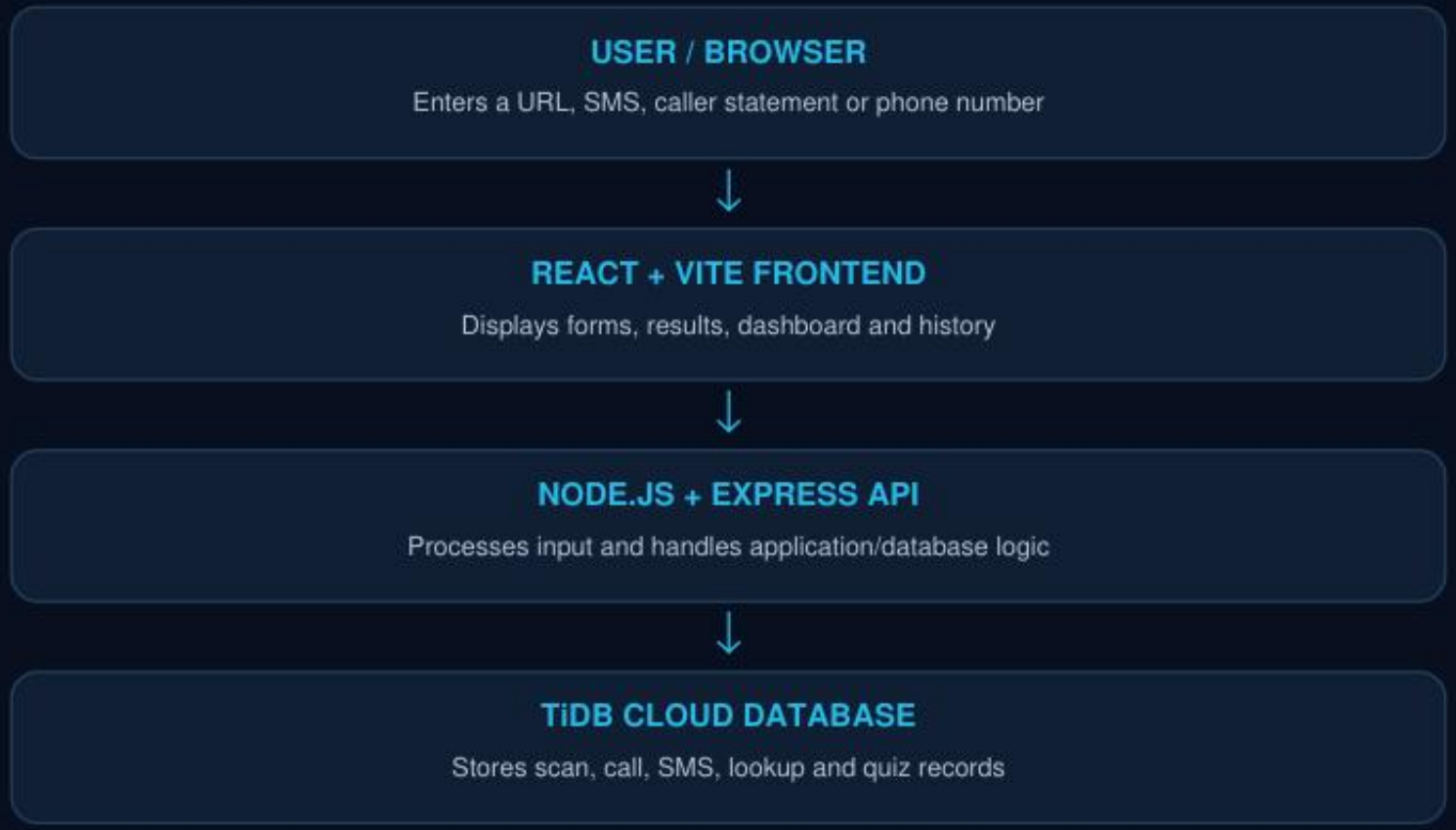
Node.js and Express.js receive requests, apply checking logic and return results.

DATABASE + HOSTING

TiDB Cloud (MySQL-compatible) stores records. Vercel hosts the frontend and backend.

How the application is connected

Requests travel from the user's browser through the backend API to the cloud database.



TiDB Cloud database

The project database is named phishguard1_db and contains six application tables.

scan_history

Stores previous URL scans, scores and classifications.

call_checks + sms_checks

Store the results of caller-statement and SMS analysis.

PHONE + QUIZ TABLES

phone_directory, phone_lookup_history and quiz_history support number references and quiz records.

Dashboard & scan analytics

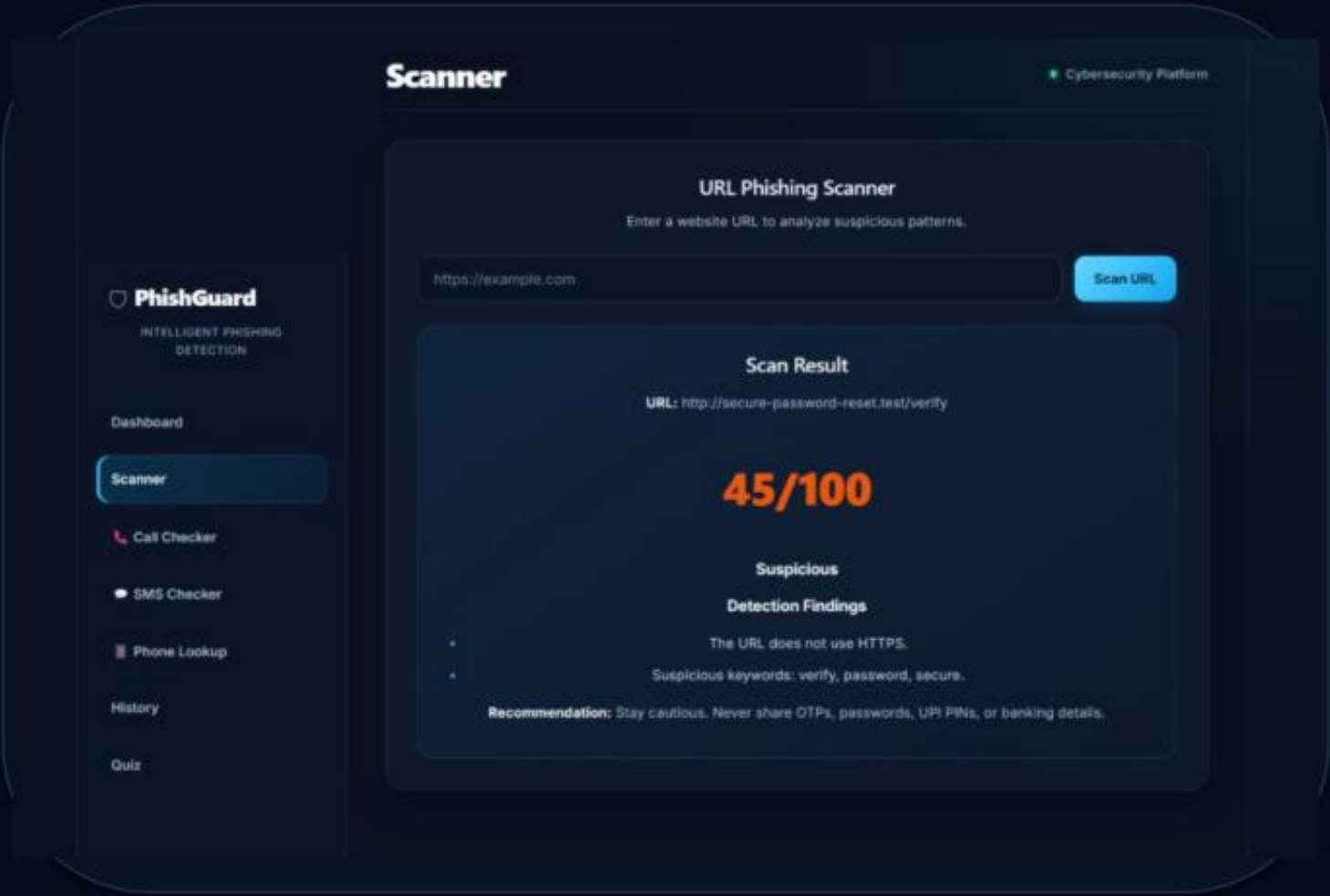


FEATURE WALKTHROUGH

A quick view of platform activity

- Displays total scans and risk-category counts.
- Visualizes the distribution of scanned URLs.
- Tracks quiz attempts and score trends.

URL phishing scanner

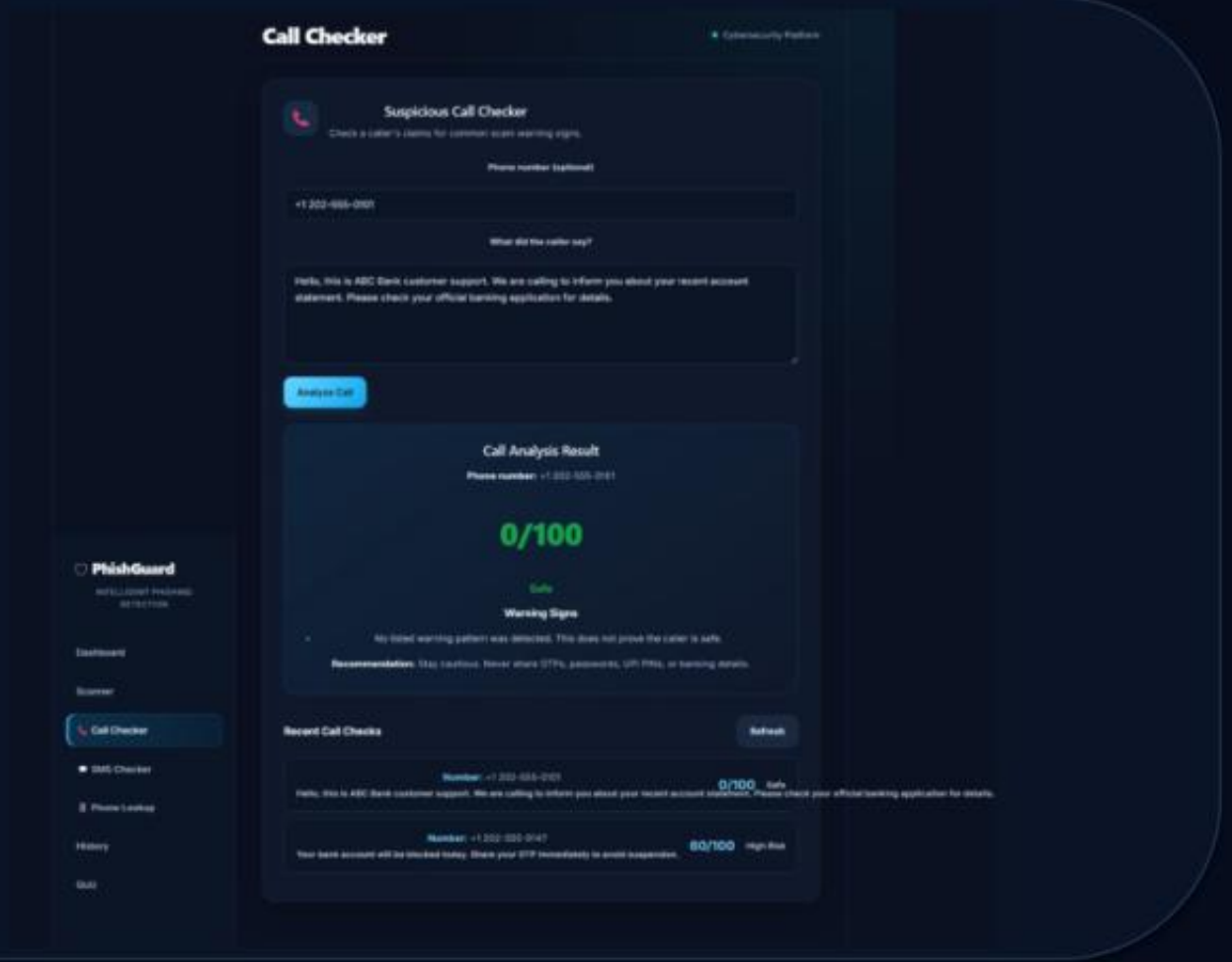


FEATURE WALKTHROUGH

Inspect a suspicious website link

- Accepts a URL for analysis.
- Shows a risk score and classification.
- Explains detected indicators and safety advice.

Suspicious call checker

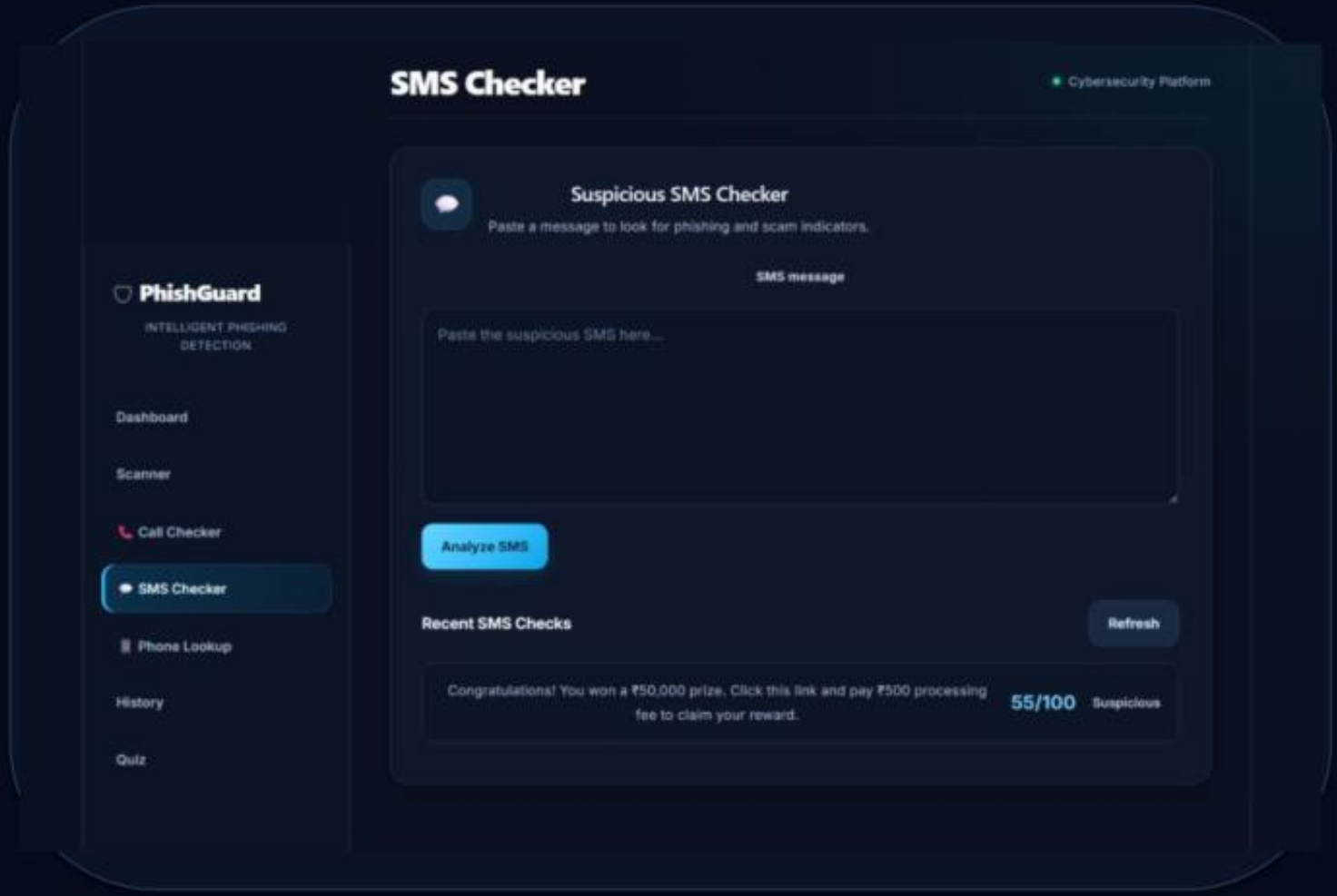


FEATURE WALKTHROUGH

Review a caller's claims

- Optional phone number and caller statement fields.
- Returns a score and warning-sign summary.
- Shows recent call checks for reference.

Suspicious SMS checker

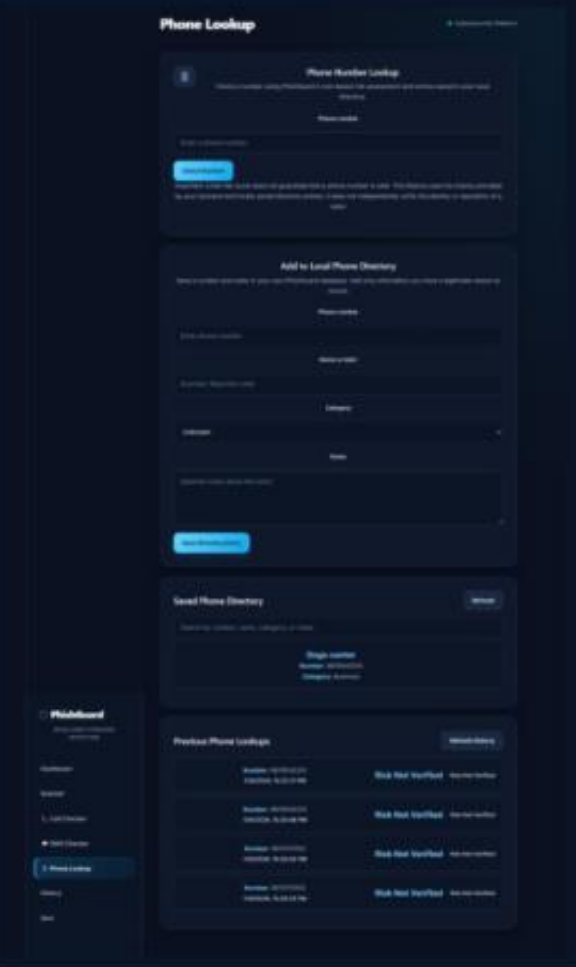


FEATURE WALKTHROUGH

Analyze suspicious message content

- Paste message text into the analyzer.
- Highlights potential phishing or scam indicators.
- Recent SMS checks can be reviewed.

Phone number lookup & directory

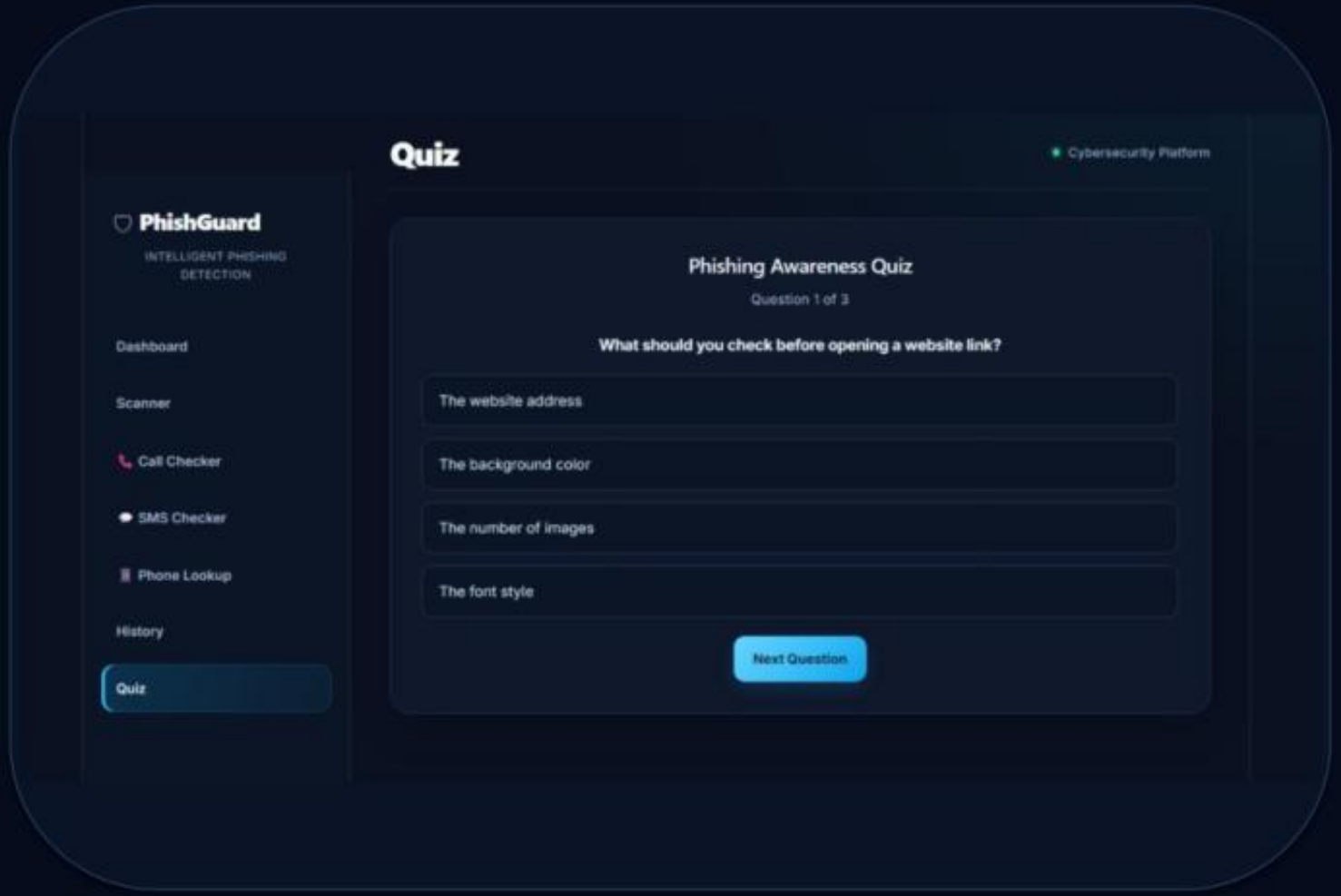


FEATURE WALKTHROUGH

Maintain a local phone reference

- Check a number using the saved directory.
- Add labels, categories and notes for known entries.
- Review previous lookup activity.

Phishing awareness quiz

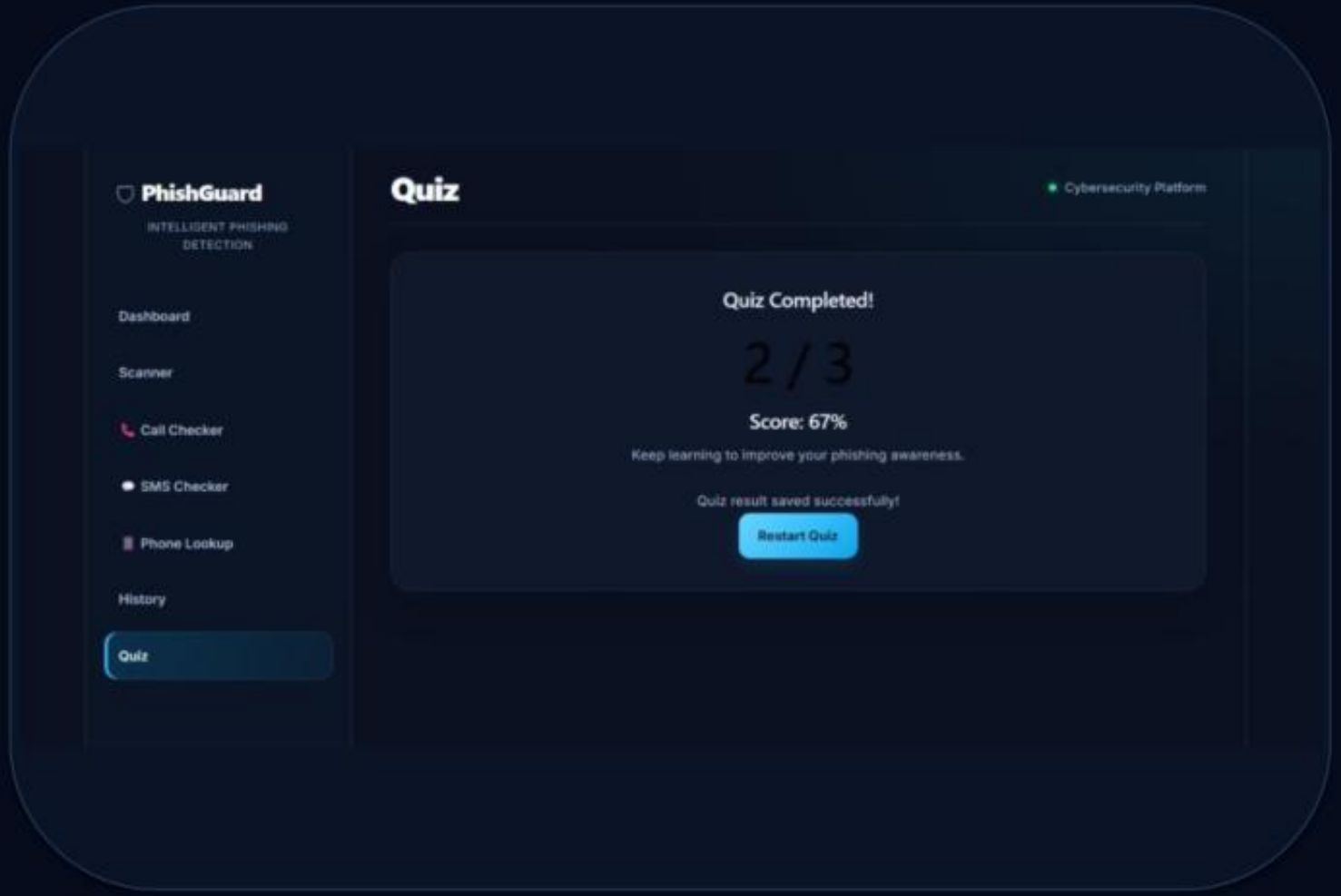


FEATURE WALKTHROUGH

Turn security awareness into practice

- Presents scenario-based questions.
- Encourages users to identify safer choices.
- Simple question flow supports self-learning.

Quiz completion & feedback



FEATURE WALKTHROUGH

Feedback after the quiz

- Displays the completed quiz score.
- Confirms when the result has been saved.
- Allows users to restart and practise again.

How Phish Guard works

01

Choose a tool

Select URL, call, SMS, phone lookup or quiz.

→

02

Enter information

Provide the URL, message, caller statement or number.

→

03

Review the analysis

Read the score, category and available findings.

→

04

Take precautions

Verify independently and avoid sharing sensitive details.

Designed to make security checks understandable and accessible to everyday users.

Testing notes, limitations & safe use

WHAT THE DEMO SHOWS

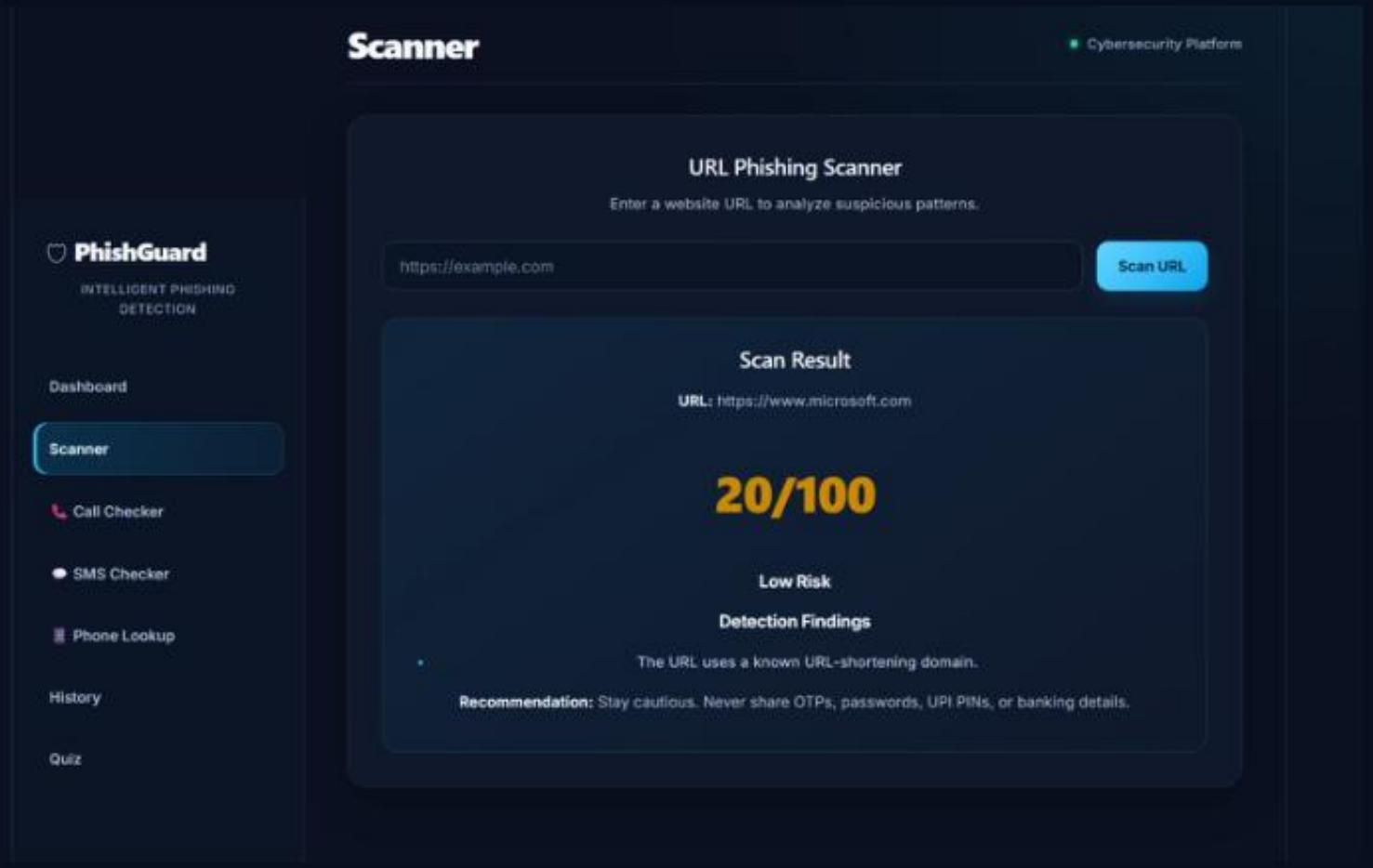
- URL scan result with a 45/100 suspicious classification in the supplied example.
- Call analysis output with a 0/100 safe classification in the supplied example.
- SMS, phone lookup and quiz interfaces are available in the screenshots.

IMPORTANT LIMITATIONS

A low score does not prove that a link, caller or phone number is safe.

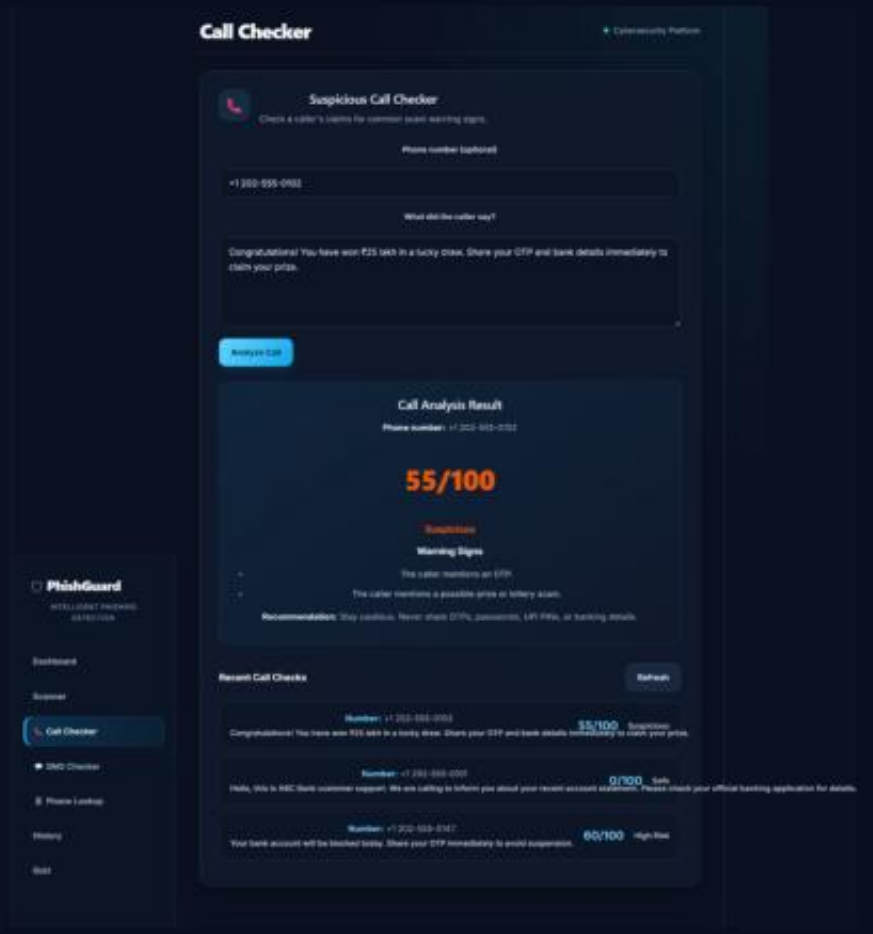
- New or changing scam patterns may not be detected.
- Verify unexpected requests using official channels.
- Never share OTPs, passwords, UPI PINs or banking details.

URL PHISHING SCANNER



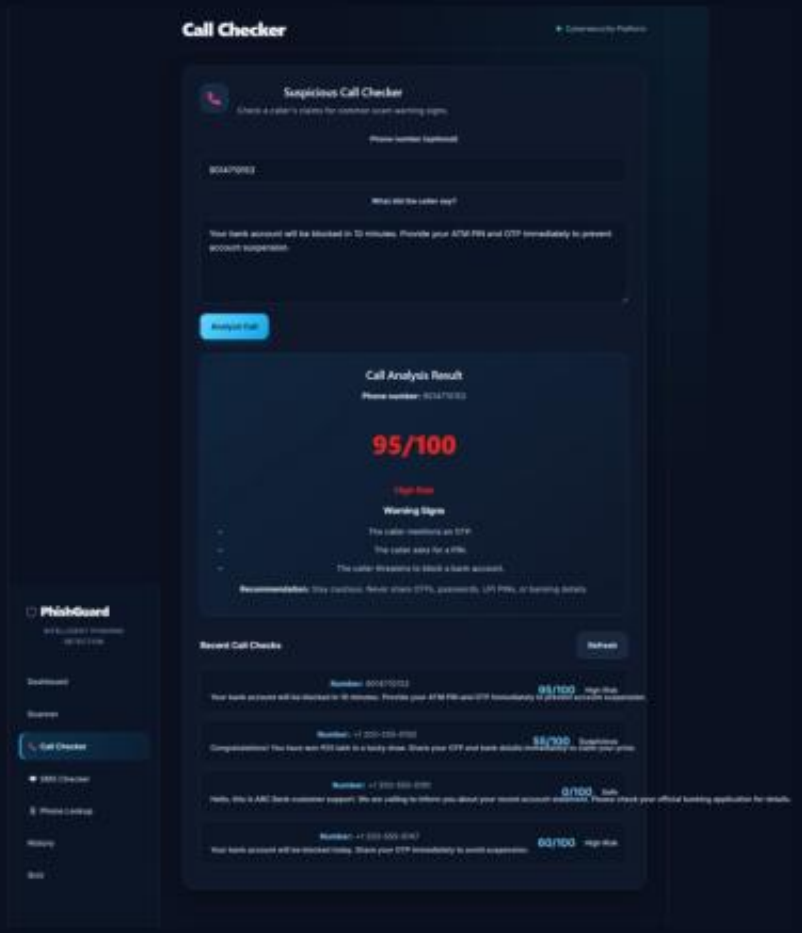
Example URL scan: 20/100 — Low Risk. The result reflects the application's configured warning patterns and is not a guarantee of safety.

CALL CHECKER — PRIZE / OTP EXAMPLE



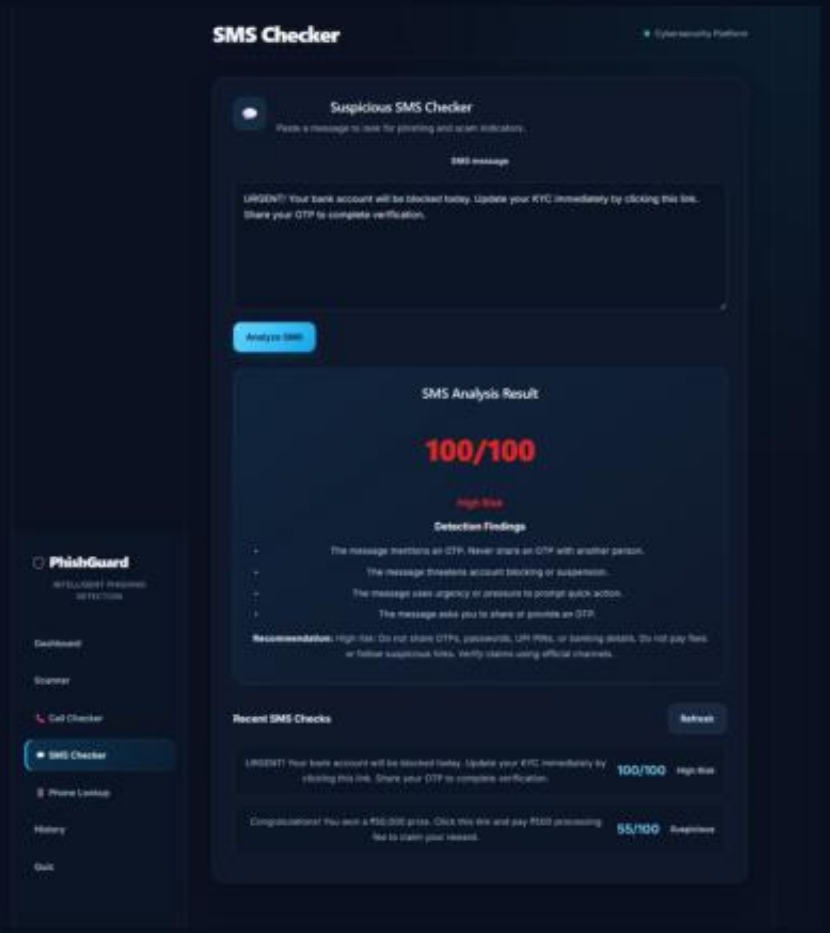
The sample caller mentions a prize and requests an OTP. PhishGuard highlights these as warning signs and returns 55/100 — Suspicious.

CALL CHECKER — ACCOUNT THREAT EXAMPLE



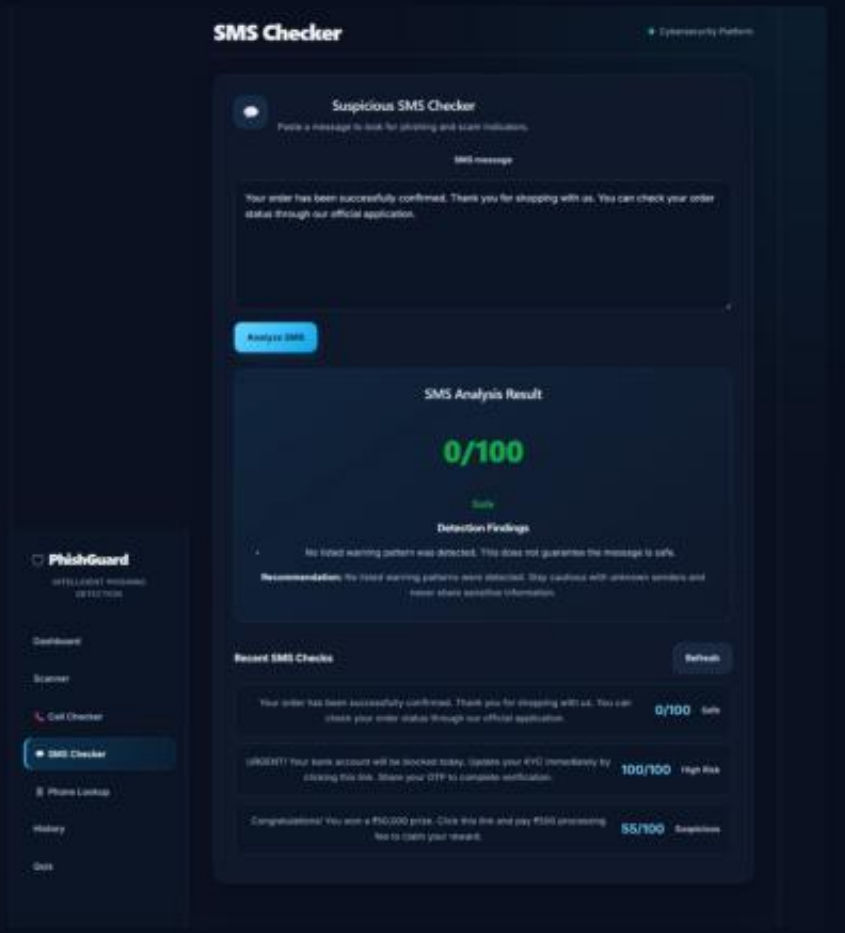
The sample caller threatens account suspension and requests an ATM PIN and OTP. The example returns 95/100 — High Risk.

SMS CHECKER — HIGH-RISK EXAMPLE



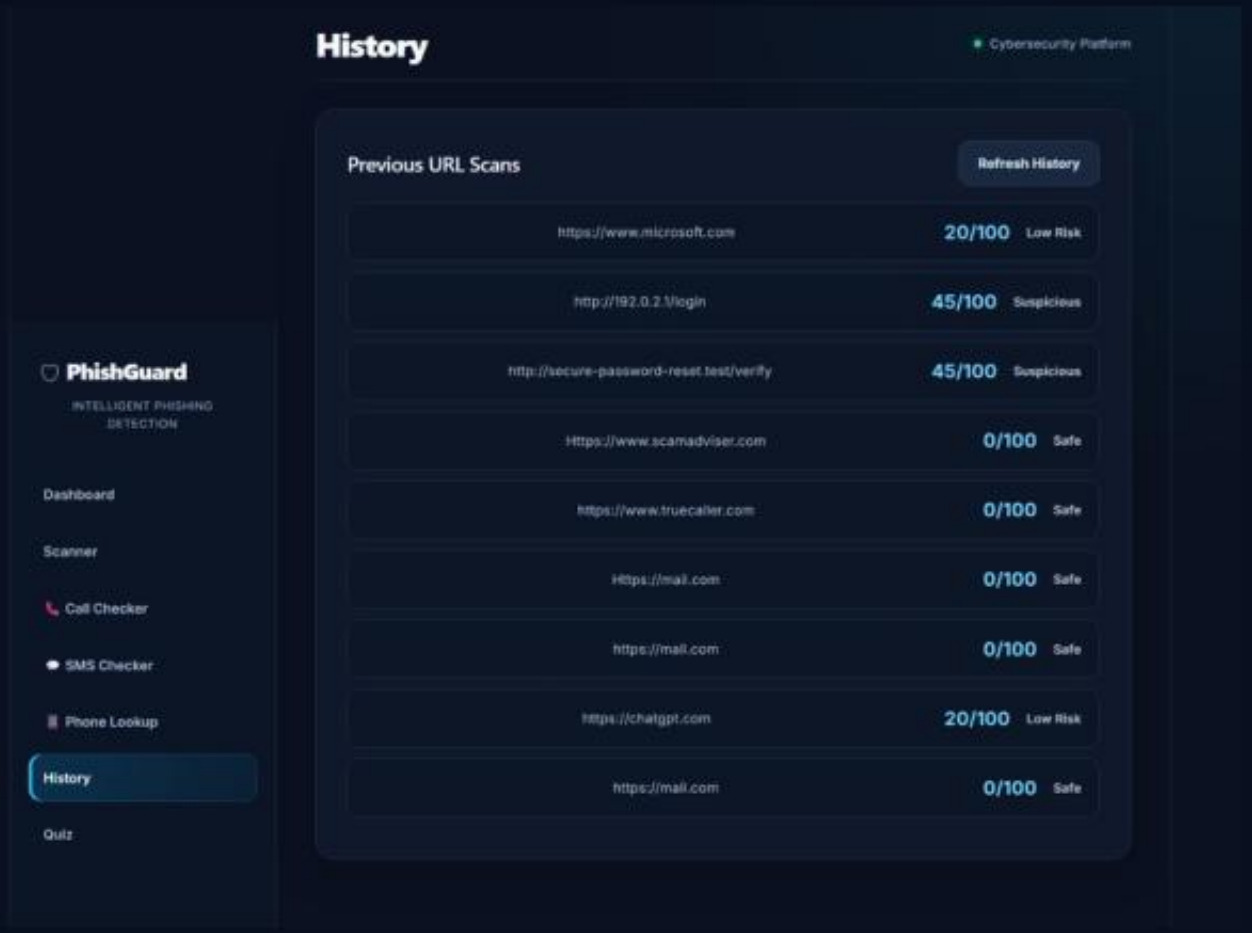
The sample SMS uses urgency, an account-blocking threat, a link and an OTP request. The example returns 100/100 — High Risk.

SMS CHECKER — ROUTINE MESSAGE EXAMPLE



No listed warning pattern was detected in this sample (0/100 — Safe). This does not prove that a message is genuine.

HISTORY — PREVIOUS URL SCANS



The History page lists previous URLs, scores and classifications so users can review earlier scan activity.

Pause before you click.

A practical cybersecurity awareness platform for checking suspicious links, calls and messages — and helping users build safer digital habits.



Live demo: phish-guard-rho-green.vercel.app

Questions & discussion